Discussion: Option-Implied Elasticity of Intertemporal Substitution

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Summary

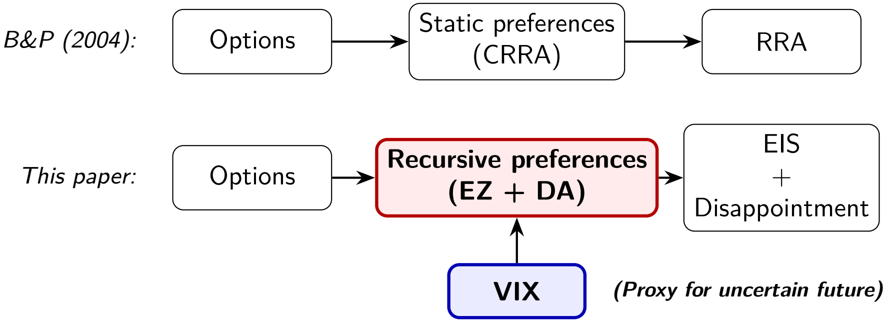
Background: EIS governs how much investors trade off today's vs. tomorrow's consumption.

Tension: Macro asset-pricing models calibrate EIS > 1. Micro and survey evidence consistently find EIS < 1.

Question: Can we estimate EIS directly from option prices?

Answer: Yes, below 1 throughout 2015–2024.

Stylized approach



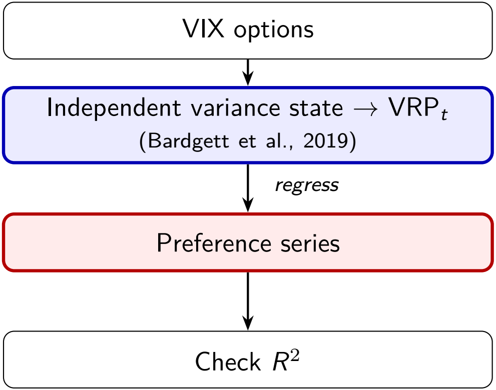
- VIX proxies the penalty the agent applies to future wealth for being uncertain.
- Interesting and ambitious, learned a lot!

Comment 1 (1/2): variance risk premium

Thought experiment

- Suppose the agent has *constant* risk preferences.
- VRP is time-varying $\Rightarrow$ VIX co-moves with it.
- SDF inherits VRP dynamics.
- Filter rationalizes this as *time-varying* preferences.

Comment 1 (2/2): second opinion with VIX options



- Association, not identification.

Comment 2 (1/2): log-utility

Log utility (RRA=1) is the price of using the VIX.

- The log-contract is one of infinitely many option-replicable payoffs; power contracts $\Rightarrow$ any fixed RRA.

Proposed solution

- Robustness: re-estimate with a power contract pinning a different RRA.

Comment 2 (2/2): example

Log enters only through the risk aggregator.

$$\mu(\mathbb{Q}_t) = \exp\big(E_t^{\mathbb{Q}}[\log S_T]\big) \quad (\text{RRA} = 1)$$

$$\Big\downarrow \; u_{risk} = x^{1-\gamma}/(1-\gamma)$$

$$\mu(\mathbb{Q}_t) = \big(E_t^{\mathbb{Q}}[S_T^{1-\gamma}]\big)^{1/(1-\gamma)} \quad (\text{RRA} = \gamma)$$

- Swap the log-contract for a power contract; same option cross-section.
- Carr–Madan: $E_t^{\mathbb{Q}}[S_T^{1-\gamma}] = \int_0^\infty w_\gamma(K)\,\mathcal{O}_t(K)\,dK$

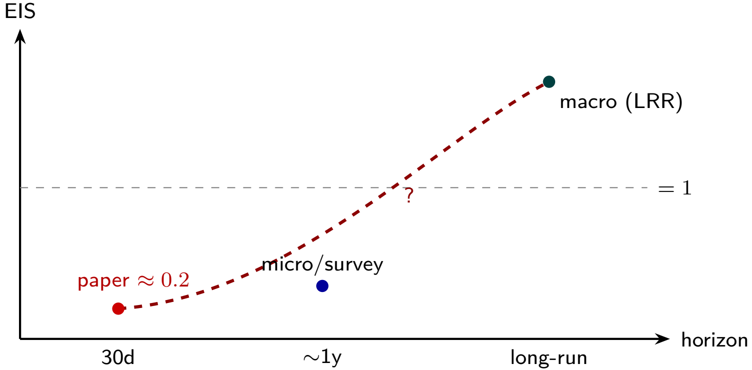
Comment 3 (1/2): which EIS are we estimating?

One parameter ψ, identified off three different margins.

- Macro: Identifies EIS off the substitution channel; needs > 1 to fit asset prices.
- Micro: Same channel, survey data; gives < 1.
- This paper: 30-day options, consumption = index. Substitution channel is less prevalent, EIS is pinned by the risk-free-rate normalization.

$\Rightarrow$ This estimate is identified off a different one.

Comment 3 (2/2): a horizon-dependence reading



- VIX3M, VIX1Y, LEAPS, etc.

Minor comments

- Sample window. 2015–2024 dominated by one crash. Extend backward using monthly-expiration contracts.
- Horizon as falsification. Re-estimate at 90 days using VIX3M or a custom log-contract.
- Density interpolation. Linear in densities (heuristic) vs. standard vol-surface interpolation.
- dP is a backward-looking KDE on few returns; tail-weighted loss and GDA lean on a tail it can't support. Estimate the density from years of overlapping monthly returns.
- Block bootstrap. Looking forward!

Conclusion

- Solutions to economic puzzles are rarely resolutions; they are often valuable reframings.
- Good luck with publication!

Thank You!